

FIELDCELL · DESIGN PRECIS

FieldCell design precis

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DOCUMENT CONTROL

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REVISION HISTORY

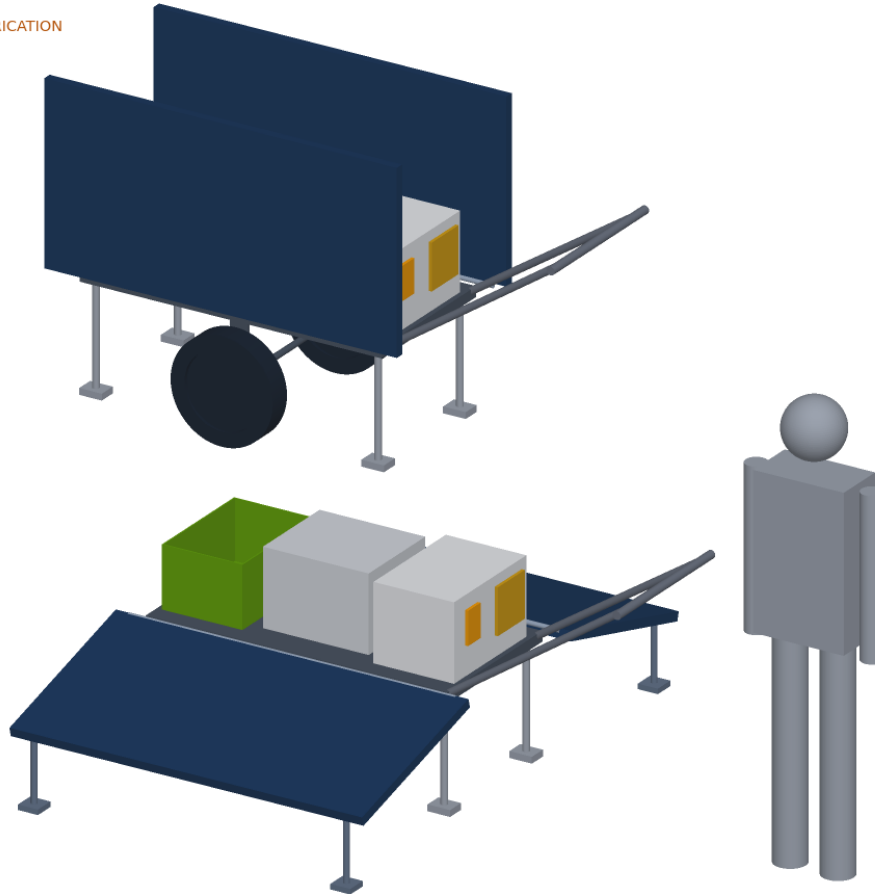
Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

FieldCell design precis

FieldCell is a two-wheel hand cart whose two sides are 200 W solar panels. For travel the panels stand up as the cart walls; on site they fold out as wings on outrigger legs. Between them sit a 1.28 kWh LiFePO₄ pack, an MPPT charger, a 1 kW inverter and DC outlets. First-order numbers suggest it stores about 1.1 to 1.4 kWh a day from its own PV at 4 to 5 peak sun hours, carries a 0.9 kWh/day field load through one day without sun, weighs about 68 kg, and deploys in about 6 minutes, for about \$1,460 in parts.

FieldCell

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. FieldCell deployed (front) and stowed for travel (behind), with a 1.75 m person for scale. Massing model.

■ How it works

1. **Travel.** The wings are latched upright as the cart sides, over the battery and electronics. One person pulls or pushes the cart by the T-handle on two 16 in (406 mm) wheels. The center of gravity sits about 60 mm on the handle side of the axle, so the handle carries a light, positive load.
2. **Park.** Four stand legs fold down so the deck sits level at about 460 mm.
3. **Deploy.** Each wing unlatches and swings out on a continuous hinge along the side rail, coming to rest 15 degrees below horizontal on two fold-down outrigger legs. With the cart's long axis north to south, one wing faces east and one faces west. In wind, the four outrigger feet are staked.

4. **Charge.** The panels are pre-wired in parallel to the MPPT charge controller, which charges the 25.6 V pack. No cable is connected on site.
5. **Power.** The operator switches on the battery isolator and the inverter. Loads plug into the AC outlet (GFCI protected) and the DC panel (USB-C PD and 12 V sockets) on the handle end of the electronics box. A shunt monitor shows state of charge.

FieldCell: daily energy flow at 4.5 peak sun hours (all values estimates)

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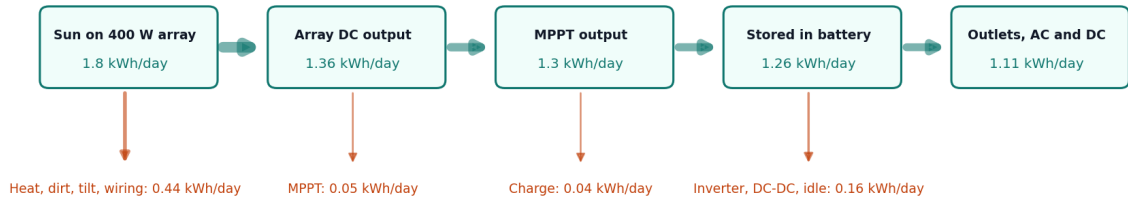


Figure 2. Daily energy flow at 4.5 peak sun hours. All values are estimates.

■ Main components

Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

Table 1. Main components.

#	Component	Proposed choice	Notes
1	Cart frame and axle	40 x 40 mm steel square tube, 1,200 x 600 mm, mesh deck, 20 mm axle	Welded; steel versus aluminium proposed, awaiting Amish
2	Wheels (2)	16 in (406 mm) flat-free or pneumatic, 720 mm track	Flat-free recommended for debris, awaiting Amish
3	Handle	Steel T-handle, grip at about 900 mm	Detaches for vehicle transport
4	Stand legs (4)	Folding tube legs with feet	Level the deck when parked
5	Battery enclosure	IP65 case, about 420 x 360 x 300 mm	Sits over the axle, lowest heavy item
6	LiFePO4 pack	25.6 V 50 Ah (1.28 kWh), 100 A BMS with low-temperature charge cutoff	24 V system proposed, awaiting Amish
7	Electronics enclosure	IP54 with filtered, baffled vent, about 320 x 460 x 300 mm	Outlet face toward the handle
8	MPPT charge controller	100 V, 20 A	About 15 A peak from 400 W
9	Inverter	1 kW continuous, 2 kW surge, pure sine, 24 V input	120 V first, 230 V variant; awaiting Amish
10	DC outlet panel	2 x USB-C PD 100 W, 2 x 12 V sockets, 24 to 12 V 20 A converter	About 440 W DC in total
11	AC outlet with GFCI	GFCI duplex (120 V) or 30 mA RCD socket (230 V), in-use cover	
12	Fusing and disconnect	Class T 100 A main fuse at the battery, isolator, DC breakers, shunt monitor	
13	PV panels (2)	200 W lightweight, about 1,400 x 700 x 35 mm, 24 V class (Vmp about 36 V)	Parallel wiring proposed, awaiting Amish
14	Panel hinges and latches	Continuous hinge per side, over-center latches	Pinch points; see Safety
15	Outrigger legs (4)	Folding legs with stake loops, set the 15 degree tilt	Stakes required above about 9 m/s wind
16	Accessory and cable bin	Open bin, balances the electronics box	Cords, lights, stakes
17	Wiring and hardware	16 mm ² (6 AWG) battery cable, PV cable, glands, fasteners	Not modeled

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3. They are computed in `cad/src/concept_media.py`.

Table 2. First-order numbers.

Quantity	Estimate	Basis	Requirement
Stored energy	1.28 kWh nominal, about 1.15 kWh usable	25.6 V x 50 Ah; 90 % depth of discharge	R1 met
PV energy into battery	1.12 to 1.40 kWh/day	400 W x 4 to 5 peak sun hours x 0.754 array derating x 0.96 MPPT x 0.97 charge	R3 met, thin at 4 h
Peak charge power	about 300 W	400 W x about 0.8 at noon, east-west wings	
Full recharge, 10 to 100 %, no load	about 0.8 day at 5 h, 0.9 day at 4.5 h	1.15 kWh / daily yield	R3 met
Reference load from battery	about 1.03 kWh/day	0.9 kWh/day at the outlets, conversion losses below	
Autonomy with no sun	about 1.1 days	1.15 kWh / 1.03 kWh/day	R4 met, thin
Battery current at 1 kW AC	about 46 A (93 A surge)	1,000 W / 0.90 / 24 V	Sets 100 A BMS and fuse
Total mass	about 68.5 kg	Table 5	R6 met, thin
Center of gravity	about 63 mm toward the handle from the axle, 575 mm above ground	Table 5	R6
Handle force, travel	about 34 N level; 17 to 52 N at 5 degree pitch	CG offset x weight / 1,240 mm axle-to-grip	R6 met
Lateral static tip angle	about 32 degrees	atan(360 mm half-track / 575 mm CG height)	R7 met
Pull force, 10 % grade	about 110 N (gravel) to 135 N (grass)	672 N weight x (0.10 grade + 0.06 to 0.10 rolling)	R7 met, thin
Ground clearance	about 190 mm	Under the axle, legs folded	R7 met
Deploy time	about 6 min	Table 6	R5 met
Deployed footprint	about 2.0 x 2.0 m including the handle	Wings at 15 degrees, 2.0 m span	
Wing lift-off wind speed, unstaked	about 9 m/s	Normal force coefficient 1.2 on 0.98 m ² , panel weight 64 N	R10 met only with stakes
Parts cost	about \$1,462	Indicative prices, bom/bom.csv	R12 met, no contingency

PV yield

The array derating of 0.754 is the product of four assumed factors: cell temperature 0.90, soiling and mismatch 0.95, wiring 0.98, and 0.90 for the east-west wings at 15 degrees compared with one panel aimed at the sun. At a mid-value of 4.5 peak sun hours the 400 W array sees 1.80 kWh/day at its rating, delivers about 1.36 kWh at its terminals, 1.30 kWh after the MPPT and 1.26 kWh into the battery (Figure 2).

Load profile and days of autonomy

The reference load is a small field post: a responder team's comms and lighting, one laptop station, and occasional tool use.

Table 3. Reference load profile (assumed).

Load	Power	Hours per day	Energy at outlet	Path
LED area lighting	30 W	8	240 Wh	DC
Radio, phone and satellite messenger charging	30 W	8	240 Wh	DC
Laptop and Wi-Fi router	50 W	6	300 Wh	AC
Power tool or small pump	700 W	1/6 (10 min)	117 Wh	AC
Total			897 Wh	

Table 4. Energy drawn from the battery (assumed efficiencies).

Path	At outlet	Efficiency	From battery
DC (USB-C and 12 V converters)	480 Wh	0.95	505 Wh
AC (inverter)	417 Wh	0.90	463 Wh
Inverter idle, 8 W for 7 h			56 Wh
Total	897 Wh	0.875 overall	1,025 Wh

With 1.15 kWh usable, the pack carries this load for about 1.1 days with no sun. At 4 peak sun hours the PV adds 1.12 kWh/day, so the system is energy-neutral with about 9 % surplus; at 4.5 h the surplus is about 23 %. A run of cloudy days is not covered. The margin is thin, which is the main technical risk in the concept.

Mass and center of gravity

Table 5. Mass budget, wings stowed, deck level. X is measured from the deck center toward the handle; the axle is at $X = 30$ mm.

Item	Mass (kg)	X (mm)	Z (mm)
Frame, deck and axle	11.0	0	440
Wheels (2)	6.0	30	203
Handle	3.0	950	700
Stand legs	1.5	0	230
Battery enclosure	3.0	0	610
LiFePO4 pack	12.0	0	570
Electronics box and contents (enclosure 3.0, inverter 4.0, MPPT 1.5, fusing and monitor 1.5, outlets 1.0)	11.0	420	590
Accessory bin and cables	4.0	-420	560
PV panels, stowed (2)	13.0	0	820
Hinges and outriggers	2.0	0	800
Wiring and hardware	2.0	200	600
Total	68.5	93	575

The center of gravity is about 63 mm on the handle side of the axle. At a grip height of about 900 mm the handle then carries about 34 N (3.5 kgf), and it stays positive between 17 and 52 N if the user holds the cart 5 degrees nose-up or nose-down. The load in the accessory bin is the trim: moving 4 kg of cords from the bin to the handle end raises the handle force by about 25 N. Mass is at the 70 kg limit of R6 with little margin; aluminium framing would save about 4 to 5 kg.

Deployment sequence

Table 6. Deploy time estimate, one person.

Step	Time
Turn the cart north to south, drop four stand legs	1.0 min
Unlatch the first wing, fold it out, swing down its two outrigger legs	1.5 min
Same for the second wing	1.5 min
Stake the four outrigger feet (when wind is expected)	1.0 min
Isolator on, check the monitor, inverter on, plug in loads	1.0 min
Total	about 6 min

■ Key design choices

- **Panels as the cart walls.** The wings protect the battery and electronics in travel and need no separate storage, cables or aiming on site. Proposed, awaiting Amish.

- **East-west wings at a fixed 15 degree tilt.** Fastest to deploy and spreads charging across the day, for an assumed 10 % yield penalty against panels aimed at the sun. Alternative: detachable panels on kickstands aimed south, about 10 to 15 % more energy for 3 to 5 extra minutes. Recommendation: east-west wings. Proposed, awaiting Amish.
- **24 V class panels in parallel.** With Vmp about 36 V each, the panels can be paralleled into one MPPT and still charge a 25.6 V pack, so the shaded or off-sun wing does not limit the other, as it would in series. Proposed, awaiting Amish.
- **25.6 V battery system.** Halves the current of a 12.8 V system (46 A instead of 90 A at 1 kW), allowing thinner cable and a smaller fuse, while 24 V inverters and MPPT controllers remain common and cheap. Alternatives: 12.8 V 100 Ah (most common parts, heavier cable) or 51.2 V. Recommendation: 25.6 V. Proposed, awaiting Amish.
- **SwapCell compatibility as a future option only.** A later variant could take a SwapCell 48 V pack in place of item 6, which would need a 48 V inverter and charger. The concept is not designed around it. Proposed, awaiting Amish.
- **Heavy parts low and over the axle.** The battery sits directly over the axle; the electronics box and the accessory bin balance each other fore and aft. Proposed, awaiting Amish.
- **Welded steel frame and flat-free tyres.** Easiest garage build and no punctures on debris, at a mass cost. Alternatives: bolted aluminium frame (lighter, costlier), pneumatic tyres (lighter, better on soft ground). Proposed, awaiting Amish.

FieldCell: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Cart frame and axle
- 2 Wheels, 16 in (2)
- 3 Handle
- 4 Stand legs (4)
- 5 Battery enclosure
- 6 LiFePO4 pack, 25.6 V 50 Ah
- 7 Electronics enclosure
- 8 MPPT charge controller
- 9 Inverter, 1 kW
- 10 DC outlet panel
- 11 AC outlet with GFCI
- 12 Fusing and disconnect
- 13 PV panels, 200 W (2)
- 14 Panel hinges and latches
- 15 Outrigger legs (4)
- 16 Accessory and cable bin

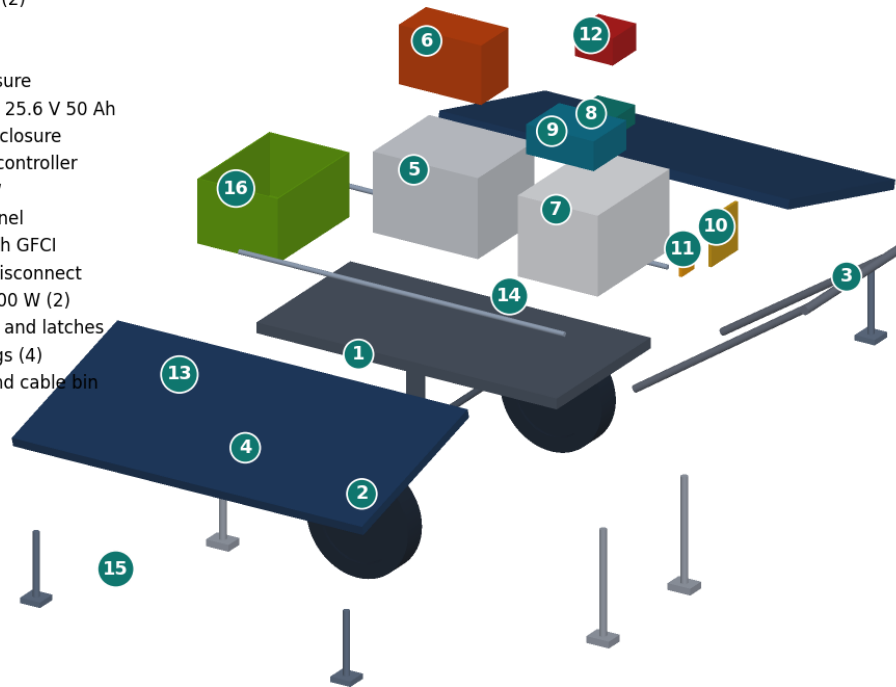


Figure 3. Exploded view with numbered callouts matching `bom/bom.csv`. Callouts 2, 4, 13 and 15 each label one of a pair; the unlabeled twin is the same part.

■ Safety

Safety: FieldCell combines a 1.28 kWh lithium battery, 120 V or 230 V AC output, high-current DC wiring, always-live PV panels and a 68 kg cart with folding wings. Every one of these is hazardous if the design or the build is wrong.

- **Lithium battery.** Use a LiFePO4 pack with a BMS that protects against overcharge, over-discharge, overcurrent, short circuit and over-temperature, and blocks charging below 0 °C. Put a class T fuse within about 150 mm of the positive terminal. Secure the pack so it cannot shift in a tip-over, keep the enclosure vented to outside air, and keep sharp objects and debris out of it. Shipping a pack of this size is regulated as dangerous goods.
- **AC output (120 V or 230 V).** Protect every AC outlet with a GFCI (120 V) or a 30 mA RCD (230 V). Bond the inverter neutral and ground to the frame only as the inverter maker specifies for a stand-alone portable source. Use in-use covers in rain, and keep outlets off wet ground.

- **Not for building wiring.** FieldCell has no power inlet and must never be connected to building or grid wiring, including by a "suicide cord" into an outlet. Back-feed can kill utility workers and damage both systems. Any such connection is out of scope.
- **DC wiring and fusing.** Fuse every circuit at its source: main class T fuse at the battery, breakers for the inverter, MPPT and DC outlets, and a PV fuse or breaker per panel string. Size cables for the fuse rating. A battery isolator must be reachable without tools.
- **PV panels are live whenever lit.** Open-circuit voltage is about 45 V per panel and cannot be switched off at the panel. Do not disconnect PV connectors under load, and cover or disconnect the panels before working in the electronics box.
- **Lifting and handling.** At about 68 kg the cart must not be lifted by one person. Use two people to load it into a vehicle, and remove the battery pack (about 12 kg) first when possible.
- **Tipping.** Lateral tip angle is about 32 degrees when stowed; do not traverse slopes across the fall line, and keep the handle low when descending. When deployed, wings can lift in wind above about 9 m/s unless the outrigger feet are staked; stow the wings in high wind.
- **Pinch points.** The wing hinges, latches and folding legs can trap fingers. Latches must hold the wings firmly in travel.
- **Heat.** The inverter and MPPT shed heat inside an enclosure that must also keep out rain. Do not cover the vents or place the unit in direct sun with the wings stowed while running at high load.

■ Open questions for TRL 3

- Check the east-west yield penalty and the temperature derating for the chosen panels with a simple irradiance model at two or three reference sites.
- Confirm the inverter idle draw and fan noise (R11) from datasheets; idle draw has a large effect on autonomy.
- Resolve the electronics box ventilation: filtered and baffled vent versus a sealed box with a heat sink wall (R8, R9).
- Wind load on the wings and the whole cart, and the stake pull-out force needed (R10).
- Handle geometry and grip height for users from 1.55 to 1.90 m tall.
- Whether an AC charger or a vehicle 12 V input should be added for cloudy periods. Proposed as an open option, awaiting Amish.
- How FieldCell relates to PowerBox and SwapCell in the portfolio, so the three do not overlap. Awaiting Amish.

Concept media: [blueprint sheet](#), [cutaway](#), [interactive 3D model](#).